

Setup

Credit Card Fraud Detection System

Prerequisites

Before running the project, make sure the following software is installed on your system.

Tool	Required Version	Check Command	Download
Python	3.9 or higher	<code>python --version</code>	https://www.python.org/downloads/
Node.js	16 or higher	<code>node --version</code>	https://nodejs.org/
npm	8 or higher	<code>npm --version</code>	Installed with Node.js
Git	Any version	<code>git --version</code>	https://git-scm.com/

These tools are required to run both the **backend server** and the **frontend application**.

Step 1 — Clone the Project

First, download the project repository to your local machine.

Open a terminal and run:

```
git clone <repository-url>
cd credit-card-fraud-detection
```

This will download the project and navigate into the main project folder.

Step 2 — Backend Setup

The backend is built using **Python, FastAPI, and SQLite**.

Open a terminal and execute the following commands.

Navigate to the backend folder

```
cd backend
```

Create a virtual environment

```
python -m venv venv
```

Activate the virtual environment

Mac / Linux

```
source venv/bin/activate
```

Windows (Command Prompt)

```
venv\Scripts\activate
```

Windows (PowerShell)

```
venv\Scripts\Activate.ps1
```

Once activated, the terminal will show `(venv)` at the beginning.

Install Python dependencies

```
pip install -r requirements.txt
```

Start the backend server

```
python main.py
```

If the backend starts successfully, you should see a message similar to:

```
INFO: Uvicorn running on http://0.0.0.0:8000
```

⚠ Keep this terminal open because the backend server must remain running.

Step 3 — Frontend Setup

The frontend is built using **React, Vite, and Tailwind CSS**.

Open a **new terminal window** and run the following commands.

Navigate to the frontend folder

```
cd frontend
```

Install Node.js dependencies

```
npm install
```

Start the frontend development server

```
npm run dev
```

If the frontend starts successfully, you will see something like:

```
VITE v5.x.x ready in xxx ms
```

```
Local: http://localhost:5173/
```

Step 4 — Open the Application

Open your web browser and visit the following URL:

<http://localhost:5173>

The application will load and you can start using the system.

How to Use the System

User Workflow

1. Register a new account from the **Register page**
 2. Login using your **email and password**
 3. Open the **Dashboard**
 4. Simulate transactions to test fraud detection
 5. View flagged transactions in the **Alerts page**
-

Admin Login (Testing)

Use the following credentials to access the admin dashboard.

Username: admin

Password: admin123

The admin dashboard allows you to:

- View all users
 - Monitor all transactions
 - View system analytics
 - Trigger high-risk test transactions
 - Retrain the machine learning model
-

Troubleshooting

Backend Module Not Found Error

Ensure that the virtual environment is activated.

You should see `(venv)` in the terminal prompt.

If not, activate it again and reinstall dependencies:

```
pip install -r requirements.txt
```

CORS Error in Browser

Make sure:

Backend is running on

```
http://localhost:8000
```

Frontend is running on

```
http://localhost:5173
```

Database Error or Missing Transactions

Delete the existing database file and restart the backend.

```
backend/fraud_detection.db
```

The system will automatically recreate the database.

npm Install Fails

Delete the following files and reinstall dependencies.

```
frontend/node_modules  
frontend/package-lock.json
```

Then run:

```
npm install
```

Port Already in Use

If port **8000** or **5173** is already in use, find the process using it and terminate it.

Example (Mac / Linux):

```
lsof -i :8000  
kill -9 <PID>
```

Project URLs (While Running)

Service	URL
Frontend Application	http://localhost:5173
Backend API	http://localhost:8000
API Documentation (Swagger)	http://localhost:8000/docs

Stopping the Application

To stop the servers, press:

```
Ctrl + C
```

in both the **backend terminal** and the **frontend terminal**.